PRAMAAN: An AyurParam-Powered Ayurveda Evidence and Safety Workbench for Classical Text Exploration and Practitioner-Assisted Review

AyushTech Innovate: National Open Ayurveda AI Challenge — Techfest, IIT Bombay 2026–27 · Phase 1 proposal Team ID: *[to fill]* · Team Leader: *[name]* · Ayurveda member (BAMS / MD (Ayu) / PhD (Ayu), NCISM-recognised institution): *[name, institution, enrolment proof uploaded? yes/no]* · Engineering members: *[names]* Repository: https://github.com/positron-nb/starDust_ayurveda (branch `pramaan-prototype`) · Licence: Apache-2.0 (code); CC BY-SA 3.0 (classical corpus, separate notice)

> Status labels used throughout: **Implemented & tested** · **Implemented, not yet validated** · **Planned**. Nothing in PRAMAAN has been clinically validated. All evaluation labels so far are synthetic and unreviewed.

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1. Executive summary

AI answers about Ayurveda are fluent but hard to verify: a reader cannot tell which sentence comes from Caraka or Vāgbhaṭa, which is the model's own background, and what was assumed about the person. PRAMAAN is **not a chatbot**; its contribution is **evidence-governed generation and auditable abstention**, not retrieval-augmented generation alone. It is a workbench in which AyurParam explains **one retrieved classical verse at a time**, the backend attaches the citation, every sentence passes citation-integrity and support checks, unsupported sentences are **withheld with a visible reason**, and a model-independent safety layer handles pregnancy, medication, emergencies and missing information. When generation fails, users still get the original verses (**source-only fallback**). A human-review workflow and an **Integrity Lab** let reviewers challenge the system. The prototype runs end-to-end with live AyurParam on an 8 GB laptop GPU.

2. Problem statement

Students and Vaidyas need to (a) locate the verse that actually addresses a question, (b) see whether an explanation is supported by that verse, and (c) avoid unsafe generalisation to a specific person. Measured on our own prototype: when given a verse, AyurParam adds content the verse does not contain in most sentences (85% of 557 generated sentences failed our automatic support check, §15). Traceability therefore has to be enforced by the system, not assumed from the model.

3. Intended users and use cases

Primary (first release): **BAMS/MD students and faculty** reviewing classical evidence on Agni, Āma and Ajīrṇa. Secondary, bounded: **Vaidyas** checking what the texts say and which safety facts are missing for an educational case. Out of scope: patients, diagnosis, prescription, dosing, safety clearance.

4. Proposed solution — workflow (Implemented & tested)

Question or synthetic case → extracted facts shown as present / absent / unknown (user can edit) → hybrid retrieval of original passages → AyurParam explains each passage → evidence ledger → safety alerts and missing-information questions → review / export packet (JSON, printable HTML). English and Hindi questions and output; Sanskrit shown as original text (IAST + derived Devanagari).

5. Innovation and novelty

We do not claim to have invented RAG or provenance. Our contribution is the combination, applied to classical Ayurveda and made inspectable: 1. **Verse-level AyurParam explanation** with server-bound citations — the model never writes citation metadata. 2. **Claim-level evidence ledger**: integrity (deterministic), support (automatic, provisional), human review (separate). 3. **Withhold-and-explain**: unsupported sentences are removed from the answer and listed with reasons. 4. **Source-only fallback** keeps the tool useful when generation fails. 5. **Model-independent safety** with explicit present/absent/**unknown** facts; unknown is never treated as no. 6. **Integrity Lab**: retract or tamper with a verse, forge a citation or inject instructions on a disposable snapshot and watch dependent claims and caches invalidate through the production validator. 7. **Honest evaluation**: "not measured" instead of placeholder numbers; displayed vs withheld populations separated. None of this guarantees factual correctness.

6. Corpus and dataset provenance (Implemented & tested)

SARIT TEI editions of the Carakasaṃhitā (Trikamji 1941 e-text) and Aṣṭāṅgahṛdayasaṃhitā (Das & Emmerick base text), pinned commit and sha256-verified; CC BY-SA 3.0 with attribution. 16,980 segments parsed; **522 indexed** for the module (Caraka Vi 2, Ci 15, Sū 5; AH Sū 8, Sū 13; + aśvagandhā mentions). Locators are the encoded `xml:id`s (e.g. AH Sūtrasthāna 13.25 = `Ah.1.13.025a–c`); no verse or page number is invented; variant readings are kept separate. Modern safety example: NCCIH aśvagandhā page (public domain), paraphrased with scope and access date. 15 English gists are unreviewed AI drafts and labelled as such. Open questions: SARIT header also says `status="restricted"`; AyurParam metadata says Apache-2.0 while its card shows CC-BY-4.0 — both recorded, not resolved.

7. AyurParam architecture and integration (Implemented & tested)

`bharatgenai/AyurParam` @ `27c69ee` (2.86 B parameters, custom `parambharatgen` Llama-style GQA) is the **only** language model at runtime. Separate model server; bf16 on GPU; custom code sha256-checked before `trust_remote_code`; 2048-token window (`config.json`) enforced with the real tokenizer; the chat template drops system messages (verified), so instructions go in the documented `<context>` block. Roles: (a) Ayurveda term expansion (few-shot, only curated concepts accepted), (b) per-passage explanation. Every run trace shows model id, revision, whether inference ran, timings and corpus/rule versions. PyTorch 2.8 tested in isolation: 8/8 outputs byte-identical to 2.7.1 (§17).

8. Hybrid retrieval design (Implemented & tested; quality is our main weakness)

BM25 over ASCII-folded IAST tokens plus curated concept tokens, and multilingual-e5-small dense vectors (IAST + curated English concept labels) in Qdrant, fused by reciprocal rank fusion; answerability gate requiring a queried domain concept with lexical support (threshold chosen on the dev split). A 20-configuration controlled study found no parameter setting that beats the default on the selection set; the failure analysis attributes misses mainly to terminology coverage, ranking among many passages sharing generic concepts, and over-broad gold labels (§15, docs/RETRIEVAL_FAILURE_ANALYSIS.md). Two targeted follow-up experiments (docs/RETRIEVAL_EXPERIMENTS.md, round 2): 64–75% of gold passages are already in the BM25∪dense candidate pool of ~44, so ordering is the main loss. A multilingual cross-encoder reranker (bge-reranker-v2-m3) improved one inspected subset but worsened the selection subset, and Roman-Hindi normalization gave only marginal gains; **neither is adopted** until a fresh, reviewer-authored locked benchmark is evaluated.

9. Evidence ledger (Implemented & tested)

Each claim stores id, text, type, evidence ids, exact support spans, source locations, interpretation status, citation integrity, support assessment, review status, limitations and producing model/prompt/ corpus versions. Results are revalidated on every display/export; a changed corpus marks them stale.

10. Citation integrity verification (Implemented & tested)

Passage exists and is not revoked; source file hash equals the manifest pin; text hash equals the hash at answer time; locator resolves; quoted spans match. Invented references ("Charaka Sūtrasthāna 45.12") and labels the model was not shown are rejected. **Integrity is structural, not semantic correctness.**

10a. Three evaluation layers (kept separate everywhere)

Layer	Question	Status
A. Citation integrity	Does the cited source exist, unchanged, at that locator?	**Measured** (57/57; 81/81)
B. Semantic claim support	Does this verse actually support this sentence?	**Not measured** — 107-record human audit prepared, labels pending
C. Clinical validity	Is the interpretation clinically appropriate?	**Not completed** — needs qualified practitioner review
A pass in A says nothing about B or C. Automatic support checks are a filter, not validation.		

11. Claim withholding (Implemented & tested; heuristic)

A sentence is shown only if its citation passes and every curated concept it mentions is present in the cited verse; dose wording, pregnancy-safety assertions and (when a safety rule is active) positive-use wording are always withheld. The check is lexical: it misses relations, negation and qualifiers, so a shown sentence can still overstate the verse — hence the human audit (§14). Withholding is a **cost** as well as a safeguard: 476/557 generated sentences were withheld and 22/60 answerable cases showed no generated sentence (the source-only fallback still shows the verses). The goal is reliable answers where evidence suffices and correct abstention where it does not — not a high withholding rate. Whether gating actually removes more unsupported than supported sentences will be measured by an **evidence-ledger ablation** on the same run (all generated sentences vs displayed ones, scored with the same human rubric) once the audit is complete; no causal benefit is claimed before that.

12. Source-only fallback (Implemented, browser-checked)

If AyurParam is unavailable or every sentence is withheld, the top verses, their gloss status and an explicit evidence-gap note are shown instead of an empty answer.

13. Safety architecture (Implemented & tested; clinical review pending)

13 YAML rules evaluated outside the model, each labelled product policy / modern safety source / classical source. Emergency wording → urgent-care template, model never called. Pregnancy, lactation, age, medication, comorbidity → sourced caution or missing-information question. A question *about* pregnancy keeps pregnancy unknown; only an explicit statement or the case box sets it present. Doses, prescriptions and advice to stop medicines are refused; unsupported herbs/interactions are "not assessed". Retrieved text is sanitised against prompt injection. Optional passcode-protected public demo mode.

14. Evaluation methodology

120 synthetic candidate cases (40 answerable, 20 multilingual, 20 safety, 20 unanswerable, 20 adversarial), grouped dev/test split. **The 77-case test split was inspected during failure analysis and is now treated as development data.** Next: (1) human claim-support audit of 107 records (57 displayed + 50 withheld) by our Ayurveda member using five labels including *unresolved*; (2) a **fresh, reviewer-authored, locked retrieval benchmark** (40–50 questions, ≥20 locked test); (3) a 5–8 student usability pilot. Tooling for all three is implemented (import is idempotent; the locked set refuses evaluation if altered).

15. Actual measured results (synthetic, unreviewed labels; not held-out)

Baseline reported on 2026-11-07 (77 former-test cases, live AyurParam, RTX 5060 Laptop):

Metric	Result
Evaluation cases / former-test cases	120 / 77
Hit@5 (≥1 relevant passage in top 5)	0.58–0.61
Passage Recall@5 (fraction of relevant passages recovered)	**0.24** (target 0.85 — not met)
Displayed claims passing citation integrity	57/57
Generated sentences withheld	316/373
Useful answers on answerable cases	23/38
Fabricated citations accepted	0/2
Emergency escalation / false alarms	4/4 · 0/73
Correct abstention on unanswerable	10/13
Latency p50	8.2 s
Human-judged claim support	**NOT MEASURED**
Clinical/domain validation	**NOT COMPLETED**

Regression run on 2026-11-09, all 120 cases (dev + inspected; a different population, not an improvement claim): citation integrity 81/81, withheld 476/557, useful answers 35/60, fabricated citations accepted 0/4, escalation 6/6 with 0/114 false alarms, abstention 17/20, p50 8.5 s; safety-rule expectations 23/23 after a stop-medication fix. Hindi/Hinglish: Devanagari close to English on a small set, Roman Hinglish clearly worse (2/7 vs 6/7 Hit@5 with the model). **Hindi and Hinglish input support is implemented experimentally, but retrieval parity has not been established.**

16. Limitations

Corpus scope: two texts, 522 passages, one theme (Agni / Āma / Ajīrṇa) — no claim of broad coverage. Weak retrieval recall; automatic support check is lexical; zero human or clinical review so far; labels synthetic; Hindi parity unproven; Suśruta, Anupāna and seasonal rules not covered; single-GPU, single-request inference; Docker untested; public hosting not yet live; licence questions above.

17. Deployment plan

(a) **Primary demo**: team laptop (RTX 5060) + passcode-protected Cloudflare tunnel exposing only the API — application controls implemented and tested, tunnel to be set up (Planned). (b) **Experiment**: inference-only Gradio Space on Hugging Face ZeroGPU behind an interchangeable adapter with automatic local fallback —

PyTorch 2.8 compatibility and the Space code verified locally; a real ZeroGPU run is Planned (requires the team's eligible HF account). (c) Docker Compose files exist; validation Planned.

18. Open-source contribution

Apache-2.0 code: secure TEI ingestion with verse locators, evidence-ledger and integrity checks, idempotent human-audit tooling, locked-benchmark tooling, Integrity Lab, evaluation runners with explicit "not measured" states. Data kept under its own licence with attribution.

19. Real-world impact (hypotheses, to be measured)

For students: faster location of the relevant verse and clearer separation of text vs interpretation. For teachers: review packets that show sources, withheld claims and open questions. We will measure task completion, time and misinterpretation in the usability pilot; no adoption, cost or revenue figure is claimed. Business hypothesis: free self-hostable core; optional institutional hosting, curated reviewed modules and support — subject to CC BY-SA terms.

20. Roadmap (dates in docs/EXECUTION_ROADMAP.md)

By 22 Nov: complete the human claim audit, terminology review, fresh benchmark authored and locked. By 1 Dec: retrieval improvements measured on the locked set, hosted demo link, usability pilot, Docker validation, updated report. After: more chapters with reviewed glosses, Suśruta, Anupāna/seasonal rules, reranking only if measured benefit.

Eligibility checklist (team to confirm — software cannot)

☐ 2–4 members · ☐ ≥1 actively enrolled BAMS / MD (Ayurveda) / PhD (Ayurveda) student from an NCISM-recognised institution · ☐ enrolment proof uploaded at registration · ☐ Team Leader designated (only they submit) · ☐ registration completed by 15 Nov.